

DEREK HUANG

U.S. Citizen

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Education

University of California San Diego

B.S. in Data Science

September 2024 – September 2026

La Jolla, California

Experience

Data Science Alliance

August 2025 – March 2026

Data Scientist Intern

Point Loma, California

- Developed Python-based data processing pipelines to ingest, clean, and transform large structured datasets for machine learning experimentation and AI-driven analytics workflows.
- Built reusable SQL queries and modular preprocessing pipelines that automated data transformation, improving reproducibility and reducing manual processing across multiple projects.
- Integrated LLM-powered summarization workflows into analytics pipelines, evaluating prompt performance through iterative testing, error analysis, and output validation to improve inference reliability.
- Collaborated with cross-functional teammates using Git-based development workflows to prototype AI-enabled data products, validate model outputs, and deliver production-ready analytical solutions.
- Improved data quality by identifying schema inconsistencies, implementing validation checks, and documenting transformation logic to support reliable downstream machine learning applications.

Delta Rising Foundation

August 2025 – December 2025

Data Scientist Researcher Volunteer

Garden Grove, California

- Designed Python workflows to collect, clean, and integrate large geospatial and environmental datasets used to identify optimal locations for redwood reforestation and carbon sequestration.
- Developed reusable preprocessing scripts that automated spatial data preparation, improving scalability and consistency across the environmental analytics pipeline.
- Applied machine learning, statistical analysis, and feature engineering to evaluate environmental variables and generate data-driven recommendations for sustainable land management.
- Collaborated with multidisciplinary teams using Git and reproducible notebook workflows to deliver technical documentation, visualizations, and validated analytical software.

Projects

LLM Math Reasoning | *Python, PyTorch, Hugging Face, Transformers, Qwen3, Prompt Engineering*

2026

- Engineered an end-to-end LLM experimentation framework using PyTorch and Hugging Face Transformers to benchmark Qwen3-4B across multiple mathematical reasoning datasets, implementing reproducible evaluation pipelines and automated inference workflows.
- Fine-tuned Qwen3-4B using Supervised Fine-Tuning (SFT), merged LoRA adapters into deployable model weights, and published the resulting model as an open-source artifact demonstrating end-to-end GenAI development.
- Evaluated LLM reasoning performance by analyzing prompt engineering strategies, decoding configurations, and model failure modes to improve inference reliability through data-driven experimentation.

BOMB Semiconductor ELT Pipeline | *Python, Apache Airflow, dbt, Docker, AWS S3, SQL, Parquet*

2026

- Engineered an end-to-end ELT pipeline that ingested **96,600** semiconductor transistor characterization records from UC Berkeley's BOMB dataset into AWS S3 using Python, Apache Airflow, Docker, and automated orchestration workflows.
- Developed modular dbt transformation models that converted raw HDF5 files into analytics-ready Parquet datasets and dimensional star-schema tables with automated data quality testing and schema validation.
- Designed scalable data models and batch processing workflows supporting semiconductor thermal stability analysis across process corners, temperature conditions, and Monte Carlo simulations while optimizing analytical query performance.

Technical Skills

Languages: Python, SQL (MySQL, PostgreSQL, BigQuery), R, Bash, JavaScript, Java, C++, C

Machine Learning & Statistics: Classification, Regression, Model Validation, Evaluation Metrics, Feature Engineering, Predictive Modeling, Hypothesis Testing, A/B Testing, Experimental Design, Time-Series Analysis

NLP & GenAI: Natural Language Processing (NLP), Unstructured Text Analysis, Large Language Models (LLMs), Generative AI, LLM Prompting, RAG Experimentation, Behavioral Signal Extraction

Data Engineering: ETL/ELT Pipelines, Data Profiling, Data Validation, Workflow Automation, Query Optimization, EDA, Cohort Analysis, KPI Development

Tools & Platforms: Linux/Unix, Git, AWS, GCP, Apache Spark, Airflow, Excel, Google Sheets, VS Code